Inverse-Root Support, Polynomial Amplitudes, and Laurent Moment Channels

All-degree low-rank coefficient atlas and persistent-moment classification

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Abstract

For a complex polynomial phase P, the cyclic rank generated by the fixed seed dx equals the nonzero residue support of the normalised inverse series at infinity and the reduced constant span of the generic inverse roots. This paper gives the unified all-degree theory in ranks at most three. Relative to named classifications of primitive polynomial monodromy and finite projective linear groups, the coefficient families are exactly powers, Dickson polynomials, quadratic- and cubic-over-power phases, indecomposable quartics, and one degree-twelve quartic collision. Every proper rank-preserving extension is also classified: power and Dickson towers, one aligned square, and one quartic collision. The proof includes algebraic finite-pole descent, a full-block stabiliser interface, a coefficient-stratified repair of the even-exponent Kummer exception, and a direct invariant-field treatment of arbitrary Dickson chains.

The fixed seed is then replaced by every polynomial amplitude $A(x)\,dx$. If $G' = A$, its cyclic rank is the number of non-trace residue channels in $G(\xi(t))$, equivalently the reduced constant span of the values $G(x_j)$. This gives the exact rank-zero condition $G \in \mathbf{C}[P]$ and a Fourier annihilation criterion for every specified zero-dimensional cycle. Combining the Pakovich–Muzychuk polynomial moment theorem with the complete low-rank right-factor list classifies persistent endpoint moments for every $q(P) \le 3$. Finally, for a Laurent phase with zero-pole order m, every Laurent one-form satisfies

$$\operatorname{Res}_0 L^k A\,dz = [\tau^{mk-1}]A(\phi(\tau))\phi'(\tau), \qquad L(\phi(\tau)) = \tau^{-m}.$$

Known indecomposable examples preclude a composition-only all-Laurent classification, so no finite global Laurent atlas is claimed. Exact replay, mutation controls, deterministic rendering, and bounded literature audits support the candidate. It remains unrefereed, not independently reproduced, fully formalised, or priority-certified.

Keywords: polynomial exponential periods; inverse-root support; cyclic rank; polynomial moments; Laurent moments; polynomial decomposition.

中文摘要

對複係數多項式相位 P，固定種子 dx 的循環秩等於無窮遠反函數級數的 非零同餘通道數，也等於一般反根的約化常數張成維數。本文統一證明循環秩不 超過三的全次數係數圖譜，分類所有保持秩的外延，並完成有限極點下降、區塊 穩定子介面、偶次 Kummer 例外修正及任意 Dickson 鏈的不變域論證。

若以任意多項式振幅 $A(x)\,dx$ 取代固定種子，且 $G' = A$，新循環秩恰為 $G(\xi(t))$ 的非跡通道數，從而得到秩零判準與指定零維循環的 Fourier 條件。結合 Pakovich–Muzychuk 多項式矩定理，可逐族分類所有 $q(P) \le 3$ 相位的恆等端點矩。對 Laurent 相位，任意 Laurent 一形式的矩 由 $[\tau^{mk-1}]A(\phi)\phi'$ 精確決定。已知不可分解例子排除單純複合型 的全域分類，故本文不宣稱有限的全 Laurent 圖譜。本成果仍是未經同儕審查的 定理候選，並非外部獨立重現、完整形式化或優先權證明。

關鍵詞： 多項式指數週期；反根支撐；循環秩；多項式矩；Laurent 矩； 多項式分解。

Combined-candidate status and lineage

> **Version 0.7.0-candidate, combined flagship manuscript.** The internal theorem and replay gates pass. The externally established theorem gate remains on hold pending unaffiliated specialist review.

This manuscript combines the complete reader-facing v0.6 all-degree paper with the polynomial-amplitude, endpoint-moment, zero-cycle, and Laurent extensions. It replaces the short v0.7 companion as the primary candidate document, but does not delete or rewrite it. The original fixed-seed papers, the v0.6 Atlas, and the companion PDF remain immutable historical artifacts.

The v0.6 correction record remains operative. Adversarial review found that a uniform full-Kummer-kernel assertion failed on an even-exponent repeated-root stratum. Sections 7–10 use the repaired coefficient-stratified theorem and preserve the exceptional aligned-square descent. The arbitrary mixed-Dickson gap is closed by the direct invariant-field proof in Section 7.

Theorem and dependency map

Result	Location	Assurance basis
fixed-seed rank, inverse support, and reduced root span	Section 2	algebraic Fourier interface; full formal-moment construction supplied in Section 14
all-degree coefficient atlas for $q(P) \leq 3$	Theorem 11.1	Sections 3–10, relative to the imported classifications listed in the references
every rank-preserving extension	Sections 7–10	corrected Kummer and direct Dickson arguments
arbitrary polynomial-amplitude rank	Theorem 14.1	formal moment and Kummer-channel proof
specified zero-cycle vanishing	Theorem 15.1	exact discrete Fourier criterion
every persistent endpoint moment with $q(P) \leq 3$	Theorem 17.1	Pakovich–Muzychuk plus Theorem 11.1
arbitrary Laurent one-form criterion	Theorem 18.1	formal residue change of variable

The all-degree proof still relies on the named primitive-monodromy and finite-projective-group classifications and on the sealed structural-repair lemmas in the source package. Those imported results are not reproved here. The amplitude and Laurent coefficient identities are proved in full in this manuscript.

Reproducibility and assurance map

The two mathematical replay gates are:

```
python3 run_global_completion_checks.py
python3 run_amplitude_laurent_checks.py
```

The reader-facing paper is rebuilt by:

```
python3 tools/build_combined_sources.py
python3 tools/render_combined_paper.py
```

The combined renderer uses Pandoc’s single-backslash mathematics extension, static KaTeX, qpdf structural validation, a content-derived document identifier, and a fail-closed scan for exposed TeX control syntax. Finite computations are regression and encoded-consequence evidence; they are not substitutes for the all-degree proofs or imported classification results.

1. Purpose and assurance boundary

The preceding successor `ALL_DEGREE_STRUCTURAL_REPAIR.md` proves the monodromy/scaling lemma and the uniform all-m Newton obstruction. Four interfaces remained:

1. the inherited equality between inverse support and reduced root span;
2. the finite-pole and infinity-order descent used on powered elementary symmetric functions;
3. the low-dimensional primitive-factor audit; and
4. the classification of rank-preserving extensions.

This note closes those interfaces in the complex polynomial setting. The primitive and extension arguments import, with exact locators, the named classification of primitive polynomial monodromy and a current public preprint on composition-length-two kernels. Those classification theorems are not reproved here. The new local arguments, the source-to-case map, and the algebraic consequences are independently checkable within this package.

The result is therefore an internally complete all-degree theorem candidate, not an externally refereed theorem and not a formalisation of the imported classification of finite groups.

Throughout, $q(P)$ denotes the fixed-seed cyclic rank, W_P the reduced constant span of the inverse roots, and $\Sigma(P)$ the nonzero inverse-at- infinity residue support after affine normalisation.

2. The inherited root-span interface

Let P be monic and centred of degree $d \geq 2$. There is a unique series

$$\xi(t) = t + O(t^{-1}) \in t\,\mathbf{C}[[t^{-1}]] \quad \text{with} \quad P(\xi(t)) = t^d. \tag{2.1}$$

Write its residue decomposition

$$X_r(t) = \frac{1}{d}\sum_{j=0}^{d-1} \zeta_d^{-jr}\,\xi(\zeta_d^j t), \qquad \Sigma(P) = \{r \bmod d : X_r \text{ nonzero}\}. \tag{2.2}$$

The inverse branches at infinity are

$$x_j = \xi(\zeta_d^j t) = \sum_{r \bmod d} \zeta_d^{jr} X_r. \tag{2.3}$$

Theorem 2.1 — Root span equals inverse support

For every complex polynomial P,

$$\boxed{\dim_{\mathbf{C}} W_P = |\Sigma(P)|.} \tag{2.4}$$

Proof

The matrices in (2.2) and (2.3) are inverse discrete Fourier matrices, so the constant span of the branches x_j equals the constant span of the channels X_r. Distinct nonzero channels are linearly independent over $\mathbf{C}$: every Laurent exponent in X_r is congruent to r modulo d, and different residue classes are disjoint. Centring gives $X_0 = 0$, which is precisely the removed trace channel. This proves (2.4). The computation takes place in $\mathbf{C}((t^{-1}))$; passing to the algebraic function field generated by the roots does not change a constant-linear dimension. □

The formal-moment construction in Theorem 14.1 below, applied to the constant amplitude, identifies the active seed channels with the same set $\Sigma(P)$. Consequently

$$\boxed{q(P) = |\Sigma(P)| = \dim_{\mathbf{C}} W_P.} \tag{2.5}$$

No analytic contour comparison is used in (2.4) or (2.5).

Corollary 2.2 — Right-factor projection

If $P = R \circ Q$, $m = \deg Q$, and $d = mn$, then

$$q(Q) = |\Sigma(P) \bmod m|. \tag{2.6}$$

Indeed, Fourier inversion within a Q-block groups the channels according to their residues modulo m. Channels with different residues have disjoint Laurent supports. This also proves heredity $q(Q) \le q(P)$. Equality holds if and only if the projection $\Sigma(P) \longrightarrow \mathbf{Z}/m\mathbf{Z}$ is injective; equivalently every generic Q-block spans all of W_P. We call this **full fusion**.

This is the only root-span input used below.

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3. Finite-pole and infinity-order descent

Let $A(Y)$ be monic of degree n, let $y_1, \dots, y_n$ be the roots of $A(Y) - z$, and choose elements v_i such that

$$v_i^m = y_i - b. \tag{3.1}$$

After the scaling lemma, monodromy acts by

$$(v_1, \dots, v_n) \longmapsto \epsilon_\gamma (v_{\sigma(1)}, \dots, v_{\sigma(n)}), \qquad \epsilon_\gamma^m = 1. \tag{3.2}$$

Put $f_k = e_k(v_1, \dots, v_n)$.

Lemma 3.1 — Invariant-integral descent

For every $1 \le k \le n$, f_k^m belongs to $\mathbf{C}[z]$. Its pole order at infinity is at most k/n.

Proof

Equation (3.2) sends f_k to
$epsilon_\gamma^k f_k$, so f_k^m is fixed by the full geometric monodromy and therefore lies in the rational base field $\mathbf{C}(z)$.

Each y_i is integral over $\mathbf{C}[z]$, being a root of the monic polynomial $A(Y) - z$. Each v_i is integral over $\mathbf{C}[z, y_i]$ by (3.1). Transitivity of integrality makes every f_k, and hence f_k^m, integral over $\mathbf{C}[z]$. Because $\mathbf{C}[z]$ is integrally closed in $\mathbf{C}(z)$,

$$f_k^m \in C(z) \cap \overline{C[z]} = C[z]. \tag{3.3}$$

At a place above infinity, $y_i = O(z^{1/n})$ and $v_i = O(z^{1/(mn)})$. Every summand of f_k is a product of k such terms, so

$$f_k^m = O(z^{k/n}). \tag{3.4}$$

This is the asserted infinity bound. □

Corollary 3.2 — Constancy below the top coefficient

If $k < n$, then f_k^m is a polynomial of integer degree at most $k/n < 1$. It is therefore constant. Since $\mathbf{C}$ is algebraically closed and the function field is an integral domain, f_k itself is constant.

For $k = n$, there is an exact identity, including its leading coefficient:

$$f_n^m = \prod_i (y_i - b) = (-1)^n (A(b) - z). \tag{3.5}$$

Thus the final coefficient is affine and nonconstant. The varying radical polynomial consequently has all lower coefficients constant and only its constant term varying. This justifies the input $B(t) + et^n$ used in the uniform coefficient extraction, without an analytic boundedness assumption.

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4. A stable line and the first rank increase

Lemma 4.1 — Stable line forces a power

If W_P contains a nonzero monodromy-stable line, then P is affine- equivalent to a power.

Proof

Let the line carry a character χ. At a chosen point over infinity, only the leading inertia character can have a pole. A stable line in any other character would have no pole over infinity and, because roots are integral over $\mathbf{C}[z]$, no finite pole; it would be constant, impossible in the reduced span. Hence $\chi(\tau) = \zeta_d$ for the full d-cycle τ at infinity.

If H is a root stabiliser, the projection of that root to the line is nonzero and fixed by H, so $H \leq \ker \chi$. Now $[G : H] = d$, while the image of χ contains an element of order d. Hence $H = \ker \chi$. The root action is faithful, so the core of H is trivial; normality gives $H = 1$ and $G = \mathrm{C}_d$. A complex polynomial with cyclic geometric monodromy is affine- equivalent to x^d. □

Therefore every nonpower root-span representation of dimension two or three is irreducible.

Choose a complete decomposition and the maximal power suffix $x^m + b$. Let A be the next indecomposable factor. Its generic roots determine distinct projective radical lines spanning a space of dimension at most three. The monodromy of A is primitive and contains a full cycle.

For completeness, the radical lines are distinct because a proportionality $u_i = \lambda u_j$ would imply $y_i - b = \lambda^m(y_j - b)$. Since a generic root is transcendental over $\mathbf{C}$, this is the polynomial identity $A(\lambda^m(Y - b) + b) = A(Y)$. A nonidentity affine deck symmetry gives a nontrivial quotient decomposition of A; the remaining identity case would make two roots equal. The maximal-power choice excludes the sole indecomposable affine-deck case.

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5. Source-exact primitive-factor audit

The audit uses two complete imported lists.

1. Theorem 3.3 of Behajaina–Koenig–Neftin [BKN26], *Monodromy groups of polynomials of composition length 2*, arXiv:2603.27609v2, states that an indecomposable complex polynomial has monodromy of one of the following types: prime cyclic or dihedral; degree-four S_4; natural S_n or odd natural A_n; or one of the explicitly listed almost-simple groups in degrees $6, 7, 8, 9, 10, 11, 13, 15, 21, 23, 31$.
2. The finite primitive subgroups of $\mathrm{PGL}_2(C)$ are $\mathrm{A}_4, \mathrm{S}_4, \mathrm{A}_5$; the primitive subgroups of $\mathrm{PGL}_3(C)$ are $\mathrm{A}_5, \mathrm{PSL}_2(7), \mathrm{A}_6, H_1, H_2, H_3$. The latter table is Table 2 of Calbeto [C25], *On the finite irreducible subgroups of GL_n(C) and PGL_n(C)*, arXiv:2510.00718v1.

The older source Burda–Khovanskii [BK16], gives the same Muller list in its theorem labelled `mullerthm`; it is retained as an independent source-to-list comparison.

Proposition 5.1 — Low-dimensional primitive audit

If the distinct radical lines belonging to the indecomposable factor A span dimension at most three, then A is affine-equivalent to

1. a prime power;
2. a prime Dickson/Chebyshev polynomial; or
3. a quartic with monodromy S_4.

Proof

In dimension two, three distinct projective points make the projective action faithful. Intersecting the PGL_2 list with the polynomial-monodromy list leaves the cyclic and dihedral families and the formal candidates S_4 in degree four and A_5 in degree five. Their point stabilisers are S_3 and A_4. A finite subgroup of $\mathrm{PGL}_2(C)$ fixing a point is cyclic, so neither candidate can stabilise a radical line. The solvable cases are precisely prime powers, prime Dickson polynomials, and degree four.

In dimension three, an irreducible imprimitive finite linear group is monomial: its diagonal subgroup is abelian and its quotient permutes three one-dimensional imprimitivity spaces through S_3. It is therefore solvable. A nonsolvable candidate must consequently act primitively. The radical lines are distinct (proportional radicals would give a constant affine relation between two distinct generic roots and hence a nontrivial block for the primitive root action), so the induced permutation action on those lines is the faithful root action. Thus every nonsolvable candidate embeds faithfully in $\mathrm{PGL}_3(C)$ and belongs to the primitive table quoted above.

The source-exact intersection of the two lists is now only

$$A_5 \text{ in degree } 5, \qquad PSL_3(2) \simeq PSL_2(7) \text{ in degree } 7. \tag{5.1}$$

For the three-dimensional A_5 character and its point stabiliser A_4,

$$\langle \chi|_{A_4}, 1\rangle = \frac{3 + 3(-1) + 8(0)}{12} = 0. \tag{5.2}$$

The two nontrivial linear characters of A_4 give the same zero because the three-cycle character value is zero. Hence A_4 fixes no projective line.

For the Klein three-dimensional representation of $\mathrm{PSL}_3(2)$ and point stabiliser S_4, the character values on identity, order two, order three, and order four are $3, -1, 0, 1$. Therefore

$$\langle \chi|_{S_4}, 1\rangle = \frac{3 + 9(-1) + 8(0) + 6(1)}{24} = 0. \tag{5.3}$$

Multiplication by the sign character also gives zero:

$$\frac{3 + 6 - 3 - 6}{24} = 0. \tag{5.4}$$

Thus S_4 fixes no projective line. Both exceptional candidates are excluded, proving the proposition. □

This expands every group, degree, stabiliser, and character used by the old one-paragraph audit. No unspecified “finite exceptional list” remains.

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6. The full-block kernel lemma

Let $P = R \circ Q$, and let $V_1, \dots, V_N$ be the spans of the generic Q-blocks, where $N = \deg R$.

Lemma 6.1 — A fixed full-span block detects the identity

Assume $q(P) = q(Q)$. Then $V_i = W_P$ for every i. If an element of the block kernel fixes one Q-block pointwise, it is the identity.

Proof

Heredity gives $\dim V_i = q(Q) = q(P) = \dim W_P$, and $V_i \leq W_P$; hence $V_i = W_P$. An element fixing the roots in one block fixes their span, so it acts identically on W_P. Every inverse root of P belongs to W_P; the element fixes all roots. The root action is faithful. □

Corollary 6.2 — Partial-support kernels are impossible

Inside the wreath-product embedding, every coordinate projection of the block kernel is injective. In particular, no nonidentity kernel element can be supported on a proper subset of the outer blocks.

This is the missing implication between full fusion and monodromy rigidity. It is stronger than the invalid shortcut “two tetrahedra span the same space, therefore they are the same tetrahedron”.

Lemma 6.3 — The full block stabiliser induces the absolute block monodromy

Let $M/\mathbf{C}(t)$ be the splitting field of $P(X) - t$, let z be one root of $R(Y) - t$, and let B be the corresponding block of roots of $Q(X) - z$. Then the setwise stabiliser S_B induces the full geometric monodromy group of Q on B.

Proof

The block system is the one attached to the intermediate field $\mathbf{C}(t, z)$. Consequently

$$S_B = \mathrm{Gal}(M/\mathbf{C}(t, z)).$$

Let L_B be the splitting field of $Q(X) - z$ over $\mathbf{C}(z)$. It is an intermediate Galois field of $M/\mathbf{C}(t, z)$. Restriction in a finite Galois extension is surjective onto the Galois group of every Galois intermediate field, so

$$S_B \twoheadrightarrow \mathrm{Gal}(L_B/\mathbf{C}(z)) = \mathrm{Mon}(Q).$$

If g carries B to another block B', then $gS_Bg^{-1} = S_{B'}$. Under rank preservation, both block spans equal W_P; hence the linear action of g on W_P conjugates the two full induced block-monodromy representations. This is the full-group conjugacy used by the normaliser arguments below, rather than a claim that a coordinate image of the outer block kernel is automatically the absolute monodromy. □

7. Rank-preserving extensions in ranks one and two

Rank one

Lemma 4.1 gives $Q = x^m$; applying it to P gives

$$P = x^{mn}, \qquad R = y^n. \tag{7.1}$$

Dickson rank two

Uniformise a generic $D_m(x, a)$ fibre by

$$x = U + a/U, \qquad W_Q = \langle U, U^{-1} \rangle. \tag{7.2}$$

If a second fibre with parameter V has the same plane, write

$$V = \alpha U + \beta U^{-1}, \qquad V^{-1} = \gamma U + \delta U^{-1}. \tag{7.3}$$

The identity $V V^{-1} = 1$ and nonconstancy of U give

$$\alpha\gamma = \beta\delta = 0. \tag{7.4}$$

Invertibility leaves $V = cU$ or $V = cU^{-1}$. We now identify the entire extension, without a prime-factor or adjacency assumption.

Let $M/\mathbf{C}(t)$ be the splitting field of $P(X) - t$. Writing the roots in the reference Q-block as

$$x_j = \zeta_m^j U + a\zeta_m^{-j} U^{-1},$$

the two Fourier projections (valid for $m \geq 3$) are

$$U = \frac{1}{m}\sum_{j=0}^{m-1} \zeta_m^{-j} x_j, \qquad \frac{a}{U} = \frac{1}{m}\sum_{j=0}^{m-1} \zeta_m^{j} x_j.$$

Thus the reference block recovers U and a/U, so $\mathbf{C}(U) \leq M$. Rank preservation makes every block span equal to $\langle U, U^{-1} \rangle$. Applying (7.3)–(7.4) to a uniformizer of every other block shows that every inverse root of P lies in $\mathbf{C}(U)$. Hence

$$M = \mathbf{C}(U). \tag{7.5}$$

Every element of $\Gamma = \mathrm{Gal}(\mathbf{C}(U)/\mathbf{C}(t))$ therefore sends U to cU or c/U. Thus Γ is a finite subgroup of the normalizer of the multiplicative torus in $\mathrm{PGL}_2(C)$. It contains the absolute block group

$$D_m = \langle U \mapsto \zeta_m U,\ U \mapsto a/U \rangle.$$

The rotational subgroup of Γ is finite cyclic, say μ_N, and the displayed reflection gives $|Gamma| = 2N$. On the other hand,

$$[\mathbf{C}(U) : \mathbf{C}(t)] = 2 \deg P = 2m \deg R = 2mn.$$

Thus $N = mn$ (and in particular $\mu_m \leq \mu_N$), and

$$\Gamma = \langle U \mapsto \zeta_N U,\ U \mapsto a/U \rangle \cong D_N. \tag{7.6}$$

Its invariant field is

$$\mathbf{C}(U)^\Gamma = \mathbf{C}\left(U^N + \frac{a^N}{U^N}\right) = \mathbf{C}(D_N(x, a)). \tag{7.7}$$

Since this field is $\mathbf{C}(t) = \mathbf{C}(P(x))$, the two generators differ by a Mobius transformation. Both are polynomials in x with their sole pole at infinity, so that transformation is affine. Consequently $P = A \circ D_N(x, a)$ for an affine A. Since $N = mn$, the Dickson identity gives

$$R \circ D_m = A \circ D_n \circ D_m.$$

Substitution by the nonconstant polynomial D_m is injective on $\mathbf{C}[Y]$, so $R = A \circ D_n$. The extension is exactly

$$D_{mn}(x, a) = D_n(D_m(x, a), a^m). \tag{7.8}$$

This argument classifies an arbitrary outer polynomial R at once. It therefore includes mixed aligned/nonaligned complete decompositions, repeated prime degrees, and every sequence of Ritt moves; no induction from a prime-degree chain theorem is required. A degree-two Dickson factor is affine-equivalent to a quadratic power and is absorbed by the power/Dickson overlap before the exact-rank-two argument is invoked.

Quadratic over a power

For $Q = (x^m + b)^2$, a block plane has a basis U, V satisfying

$$U^m + V^m = -2b, \qquad b \in \mathbf{C}^\times. \tag{7.9}$$

An equal block plane produces an invertible linear substitution preserving this level. Homogenising shows that it preserves the binary Fermat form $U^m + V^m$. For $m \geq 3$, its linear automorphism group is generated by coordinate scalings with m-th power one and interchange of the coordinates. Thus the new fibre parameter is the old one or its negative before squaring, which is the same Q-fibre. No distinct full-fusion block is created.

For $m = 2$, this family is the degree-four Dickson overlap and is already covered by (7.8). Hence Dickson chains are the only proper rank-two- preserving extensions.

8. Rank-three cubic-over-power extensions

Write

$$Q = H(x^m), \qquad H(Y) = Y^3 + AY^2 + BY + C, qquadm \geq 2. \tag{8.1}$$

Exact rank three means (A, B) is nonzero, except that $m = 2$, $B = A^2/4 \in \mathbf{C}^\times$ is the rank-two Dickson overlap. The affine-power locus $3B = A^2$ remains rank three for $m \geq 2$, but its cubic monodromy is C_3 and must be treated separately from the S_3 branch. In a generic fibre, the three Kummer blocks span independent lines with generators U_i, where the powered values $Y_i = U_i^m$ have fixed first and second elementary symmetric functions.

Lemma 8.1 — Kummer-line transport on the full-kernel strata

Off the exceptional stratum $B \in \mathbf{C}^\times$, $A^2 = 4B$, $2 \mid m$, transport between two generic Q-blocks under a rank-preserving extension has the form

$$U_i' = c_i U_{\pi(i)}. \tag{8.2}$$

Proof

By Lemma 6.3, the full setwise stabiliser of every Q-block induces the absolute block monodromy. An element carrying one block to another conjugates these full actions on the common three-dimensional span.

Off the exceptional stratum $B \in \mathbf{C}^\times$, $A^2 = 4B$, $2 \mid m$, the Kummer rotations form the full diagonal normal subgroup of the monomial block action. For $m \geq 3$, its three weight lines are the unique one-dimensional imprimitivity system and hence are preserved by every such coordinate identification. For $m = 2$, the projective block action is the octahedral S_4; its projective normaliser is itself and preserves the orbit of the three opposite-axis pairs, which are the same Kummer lines. Thus in the nonpower cases a block transport permutes the three weight lines, giving (8.2). On the affine-power locus $3B = A^2$, the projective exponent-two group is tetrahedral A_4; its normaliser is generated by octahedral signed-monomial matrices and diagonal character intertwiners, so it preserves the same axes. The spectral argument handles this locus for $m \geq 3$. $\square$

Put $a_i = c_i^m$. The powered triples Y_i and $a_i Y_i$ have the same fixed first and second symmetric functions. On the nonpower S_3 locus, the constant-linear relation space of the generic cubic roots is zero when $A \in \mathbf{C}^\times$ and is the trace line when $A = 0$. Consequently all a_i are one common scalar c. The fixed coefficients give

$$cA = A, \qquad c^2 B = B. \tag{8.3}$$

If $A \in \mathbf{C}^\times$, then $c = 1$. If $A = 0$ and $B \in \mathbf{C}^\times$, the sole nontrivial possibility is $c = -1$. Hence the complete fibre-equivalence group is at most μ_2, and equality occurs precisely when the centred cubic is odd. The only proper extension is

$$Q = x^{3m} + bx^m, \qquad P = Q^2, \qquad b \in \mathbf{C}^\times. \tag{8.4}$$

On the affine-power locus, write the roots as $Y_i = -a + \omega^i w$. Preservation of the first two elementary symmetric functions gives

$$\sum a_i = 3, \quad \sum a_i \omega^i = 0, \quad a_1^2 = a_2^2 = a_3^2 = 1.$$

Therefore every $a_i = 1$, and this locus has no nontrivial fibre quotient.

It remains to treat the exceptional stratum excluded from Lemma 8.1. Write $A = 2b$, $B = b^2$, and $m = 2k$. Then

$$H(x^{2k}) - C = (x^{3k} + bx^k)^2.$$

For $k = 1$ this is, up to target translation, D_6 and has rank two. For $k \geq 2$ it is already the aligned-square quotient of the nonexceptional phase $F = x^{3k} + bx^k$. The nonexceptional calculation just completed proves that the full rank-preserving fibre quotient of F has degree at most two. Any further nonlinear outer factor on F^2 would therefore give a quotient of F of degree greater than two, which is impossible. No transport assertion for the reduced Kummer kernel is needed on this stratum.

The extension (8.4) cannot repeat. After rewriting it as a cubic over x^{2m}, the outer cubic is

$$Y(Y + b)^2 = Y^3 + 2bY^2 + b^2Y, \tag{8.5}$$

which is the exceptional even stratum already isolated in the proof of Lemma 8.1. It is already the degree-two quotient of $x^{3m} + bx^m$; a further nonlinear quotient would compose to a rank-preserving quotient of that phase of degree greater than two, contradicting the complete nonexceptional fibre-equivalence calculation.

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9. Rank-three quartic extensions

Let

$$Q = x^4 + Ax^2 + Bx, \qquad B \in \mathbf{C}^\times. \tag{9.1}$$

Its monodromy is S_4 and its root span is the standard three-dimensional representation.

Take the rightmost indecomposable factor G of an outer extension R; then $G \circ Q$ is a right factor of $R \circ Q$. Heredity forces

$$q(Q) \leq q(G \circ Q) \leq q(R \circ Q) = 3, \tag{9.2}$$

so $G \circ Q$ is already rank preserving.

The indecomposable-monodromy list and the length-two kernel theorems of Behajaina–Koenig–Neftin [BKN26] are exhaustive:

- if G is an AGL_1 polynomial, their Theorem 2.3 says that the kernel contains $V_4^{\deg G}$ in every case except the Ritt collision $X^3 \circ ((X^3 + 1)X)$;
- if G has S_4 monodromy, their Theorem 2.4 contains V_4^4;
- if G is nonsolvable, their Theorem 2.7 contains $V_4^{\deg G}$.

Every displayed product contains a nonidentity element supported on one outer block, contradicting Corollary 6.2. Therefore only the Ritt collision can preserve rank. In coefficient form it occurs exactly when $A = 0$:

$$Q(cx) = cQ(x) \quad (c^3 = 1), \qquad P = Q^3 = (x^4 + Bx)^3. \tag{9.3}$$

There is no further extension. The projective monodromy of (9.3) is the tetrahedral S_4; its normaliser in $\mathrm{PGL}_3(C)$ is itself because every automorphism of S_4 is inner. Thus any further full-fusion transport is, after a root permutation, scalar. Coefficient comparison in

$$(x^4 + Bx)^3 = x^{12} + 3Bx^9 + 3B^2x^6 + B^3x^3 \tag{9.4}$$

shows that a scalar semiequivariance has $c^3 = 1$ and target multiplier one. It is already a deck transformation and creates no distinct fibre.

10. Completion of the first-increase cases

The primitive audit leaves prime powers, prime Dickson factors, and quartics. The power computation gives ranks two and three only for outer degrees two and three. The Dickson calculation gives the aligned half-angle identity in rank two; its only rank-three first increase is the cubic case already covered by the uniform Newton descent. The quartic case is exactly the uniform quartic descent.

By Corollary 3.2 the lower radical coefficients in those descents are constant, so the coefficient-extraction argument applies without an analytic qualification. `ALL_DEGREE_STRUCTURAL_REPAIR.md` then proves:

$$\text{cubic lift: } m = 2, \qquad \text{quartic lift: } m = 3 \text{ and } A = 0. \tag{10.1}$$

The latter is (9.3). The former is the aligned Dickson/power-square overlap.

11. All-degree theorem

Theorem 11.1 — Complex low-rank coefficient atlas

Let $Pin\mathbf{C}[x]$ have degree at least two. Up to invertible affine changes of source and target, $q(P) \le 3$ if and only if P belongs to one of the following families, with the lower-rank intersections removed in the stated rank:

1. rank one: powers x^d;
2. rank two: Dickson polynomials $D_d(x, a)$ and quadratic-over-power phases $x^{2m} + Ax^m$;
3. rank three: cubic-over-power phases $x^{3m} + Ax^{2m} + Bx^m$, indecomposable quartics $x^4 + Ax^2 + Bx$ with $B \in \mathbf{C}^\times$, and the collision $(x^4 + ax)^3$ with $a \in \mathbf{C}^\times$.

The proper rank-preserving extensions are exactly:

$$\begin{aligned} &\text{powers:} && x^m \mapsto x^{mn}, \\ &\text{Dickson:} && D_m(x, a) \mapsto D_{mn}(x, a), \\ &\text{aligned square:} && x^{3m} + bx^m \mapsto (x^{3m} + bx^m)^2, \\ &\text{quartic collision:} && x^4 + ax \mapsto (x^4 + ax)^3. \end{aligned} \tag{11.1}$$

Proof

Read ranks from the right along a complete decomposition. Rank-one suffixes combine to a power. The stable-line lemma makes every nonpower rank-two or rank-three span irreducible. The $[2, 1]$ support profile would give a common stable line, so rank cannot increase from two to three. Proposition 5.1 and the uniform descent classify the first strict increase above the maximal power suffix. Sections 7–9 classify every subsequent rank-preserving step and show that the two rank-three steps cannot repeat. These alternatives are exactly the displayed families. Direct inverse-support computations, already sealed in the coefficient atlas and degree-32 successor, establish existence and the stated lower-rank intersections. □

Combining Theorem 11.1 with (2.5) removes the last all-degree qualifier from the algebraic fixed-seed atlas.

---

12. What is and is not closed

Closed internally in this successor:

- the local root-span/support interface needed by the atlas;
- finite-pole descent and the exact infinity-degree bound;
- the complete source-to-group primitive audit;
- the missing full-block-kernel implication;
- the four rank-preserving extension mechanisms and non-repetition; and
- the resulting all-degree classification over $\mathbf{C}$, relative to the named finite-group and polynomial-monodromy classification theorems.

Not claimed:

- a reproof of the classification of primitive polynomial monodromy groups;
- a reproof of the finite-subgroup classifications of PGL_2 and PGL_3;
- formal verification of algebraic function-field integrality or those group classifications;
- external independent reproduction, specialist peer review, novelty, or publication; or
- positive-characteristic extensions.

Sections 13–19 below close the polynomial-amplitude and Laurent local-channel extensions. They preserve a separate boundary around a finite all-Laurent normal-form theorem.

---

13. Three notions of persistent vanishing

We distinguish three questions.

1. **Amplitude cyclicity.** What is the rank generated by $[A\,dx]$ under the exponential connection for a fixed polynomial phase P?
2. **Endpoint or zero-cycle moments.** For which amplitudes do all moments vanish between fixed endpoints, or on a fixed zero-dimensional cycle?

3. **Laurent residues.** For which Laurent one-forms do all residues $\operatorname{Res}_0 L^k A\,dz$ vanish?

These questions interact, but they are not identical. In particular, $q(P, dx) \le 3$ does not imply that every amplitude has cyclic rank at most three.

Let $d = \deg P \ge 2$, put $K = \mathbf{C}(s)$, and write

$$\mathcal{H}_P = K[x]\,dx/(\partial_x + sP')K[x]\,dx. \tag{13.1}$$

The connection is

$$\nabla_s[f\,dx] = [(\partial_s f + Pf)\,dx]. \tag{13.2}$$

For $A \in \mathbf{C}[x]$, define

$$q(P, A\,dx) = \dim_K K\langle\nabla_s\rangle[A\,dx]. \tag{13.3}$$

An affine source change transports both P and $A\,dx$; an affine target change acts triangularly on powers of P. We may take P monic and centred when computing support.

---

14. Primitive-amplitude support

Let

$$P(\xi(t)) = t^d, \qquad \xi(t) = t + O(t^{-1}) \tag{14.1}$$

be the normalised inverse at infinity, and let $G' = A$. Write

$$G(\xi(t)) = \sum_{n \le N} c_n t^n. \tag{14.2}$$

The **primitive-amplitude support** is

$$\Sigma(P; G) = \{r \in \{1, \dots, d-1\} : c_n \text{ is nonzero for some } n \equiv r \pmod d\}. \tag{14.3}$$

Changing the additive constant in G changes only the trace channel.

Theorem 14.1 — Polynomial-amplitude support theorem

For every polynomial phase and polynomial amplitude,

$$\boxed{q(P, A\,dx) = |\Sigma(P; G)| = \dim_{\mathbf{C}} \operatorname{span}^{\mathrm{red}}\{G(x_0), \dots, G(x_{d-1})\}.} \tag{14.4}$$

Here $x_j = \xi(\zeta_d^j t)$, and "red" removes the common trace component.

Proof

We construct the formal moment isomorphism in the form needed here. Pull a one-form back by $x = \xi(t)$. In nonzero residue class $r \pmod d$, send

$$t^{r+nd-1}dt \longmapsto \mu_{r,n} s^{-n} e_r, \qquad \mu_{r,n+1} = -\frac{r+nd}{d}\mu_{r,n}. \tag{14.5}$$

All $\mu_{r,n}$ are nonzero. Equation $P(\xi) = t^d$ makes $(\partial_x + sP')h\,dx$ pull back to $(\partial_t + sdt^{d-1})h(\xi(t))\,dt$, which maps to zero by (14.5). The map descends to

$$\mathcal{H}_P \otimes_K \mathbf{C}((s)) \longrightarrow \bigoplus_{r=1}^{d-1} \mathbf{C}((s))e_r. \tag{14.6}$$

On $dx, x\,dx, \dots, x^{d-2}dx$, its constant matrix is triangular with diagonal one because $\xi^j\xi' = t^j + O(t^{j-2})$. Hence (14.6) is an isomorphism.

Now

$$A(\xi)\xi' dt = d(G(\xi)) = \sum_n n c_n t^{n-1} dt. \tag{14.7}$$

For a nonzero residue r, n cannot vanish when $n \equiv r \pmod d$. Distinct coefficients occupy distinct powers of s, so the r-th component is nonzero exactly when $r \in \Sigma(P;G)$. For each active residue choose its leading power of s and write the corresponding exponent as alpha_r. Distinct residues give distinct alpha_r modulo the integers. The leading coefficient of the derivative Wronskian is the product of the nonzero channel coefficients and the Vandermonde product over the alpha_r. It is nonzero. The cyclic rank is therefore the number of active channels.

Finally,

$$Y_r(t) = \frac{1}{d}\sum_{j=0}^{d-1} \zeta_d^{-jr} G(x_j(t))$$

is the part of (14.2) in residue class r. Fourier inversion and disjoint Laurent supports prove the reduced-span equality. □

Corollary 14.2 — Universal rank-zero amplitudes

$$q(P, A\,dx) = 0 \quad\Longleftrightarrow\quad \Sigma(P;G) = \varnothing \quad\Longleftrightarrow\quad G \in \mathbf{C}[P]. \tag{14.8}$$

Equivalently, $A(x) = P'(x)H'(P(x))$ for some polynomial H.

Proof

If $G = H \circ P$, then $G(\xi(t)) = H(t^d)$. Conversely, empty non-trace support makes $G(x_j)$ equal on all generic inverse roots. Hence $G \in \mathbf{C}(P)$ inside $\mathbf{C}(x)$. Because both P and G are polynomials and have no finite pole, $\mathbf{C}(P) \cap \mathbf{C}[x] = \mathbf{C}[P]$. □

Example 14.3 — Low phase rank does not bound amplitude rank

For $P = (x^4 + x)^3$, $q(P, dx) = 3$. Taking $G = x^4$, $A = 4x^3$, gives

$$\Sigma(P; x^4) = \{1, 4, 7, 10\}, \qquad q(P, 4x^3 dx) = 4. \tag{14.9}$$

15. Exact zero-cycle classification

Let $n = (n_0, \dots, n_{d-1}) \in \mathbf{C}^d$, $\sum_j n_j = 0$, and put

$$\widehat{n}(r) = \sum_{j=0}^{d-1} n_j \zeta_d^{jr}. \tag{15.1}$$

Theorem 15.1 — Support annihilation on a zero-cycle

$$\sum_{j=0}^{d-1} n_j G(x_j(t)) \equiv 0 \quad \Longleftrightarrow \quad \widehat{n}(r) = 0 \text{ for every } r \in \Sigma(P; G). \tag{15.2}$$

Proof

Fourier inversion writes $G(x_j) = \sum_r \zeta_d^{jr} Y_r$. The trace term vanishes because $\widehat{n}(0) = 0$. Nonzero Y_r are constant-linearly independent because their exponents occupy disjoint residue classes. $\square$

This is complete for a specified phase, amplitude, and cycle. It does not make the broader balanced-cycle literature composition-only [ABM13].

For $P = (x^4 + x)^3$, the cycle $n_j = \cos(2\pi j/3)$ has Fourier support $\{4, 8\}$. The primitives

$$G_7 = x^7 + \frac{7}{4}x^4, \qquad G_{10} = x^{10} - \frac{5}{2}x^4 \tag{15.3}$$

both have support $\{1, 7, 10\}$, proving their zero-cycle vanishing. Their amplitudes are $7x^6 + 7x^3$ and $10x^9 - 10x^3$.

16. Endpoint moments and the imported theorem

Fix $a, b \in \mathbf{C}$, put $G' = A$, and define

$$M_k(P, A; a, b) = \int_a^b P(x)^k A(x)\, dx. \tag{16.1}$$

Theorem 16.1 — Polynomial moment theorem [PM09]

All $M_k(P, A; a, b)$ vanish for $k \geq 0$ if and only if

$$G = \sum_{j=1}^{r} H_j \circ W_j, \tag{16.2}$$

where $P = R_j \circ W_j$ and $W_j(a) = W_j(b)$ for every j.

Necessity is imported from [PM09], not reproved. Sufficiency follows by substitution $u = W_j(x)$. Pakovich's refinement [P13] describes the indecomposable right factors explicitly.

If $P(a)$ and $P(b)$ differ, no nonconstant right factor is endpoint-admissible, so the only solution is $A = 0$. Hence assume $P(a) = P(b)$.

---

17. Persistent endpoint moments for every $q(P) \le 3$

Theorem 16.1 now specialises through the all-degree normal forms of Theorem 11.1 and the right-factor audit in Sections 7–9. Because Theorem 11.1 remains unrefereed relative to named monodromy classifications, the same qualification attaches here. Formulas are in Atlas coordinates; affine source changes transport $a, b, A\,dx, G$.

Theorem 17.1 — Low-rank persistent-moment classification

Let $q(P, dx) \le 3$, let a, b be distinct, and let $P(a) = P(b)$. All endpoint moments vanish if and only if G belongs to the final-column space below.

Phase	Endpoint-admissible factors	Complete space for G
$P = x^d$	x^h, where h is the order of a/b	$\mathbf{C}[x^h]$
$P = D_d(x, \alpha)$	divisibility-minimal $D_e(x, \alpha)$ with $e \mid d$ and $D_e(a, \alpha) = D_e(b, \alpha)$	$\sum_{e \in E_{\min}} \mathbf{C}[D_e]$
$P = H(x^m)$, $\deg H = 2$	least x^h, $h \mid m$, with $a^h = b^h$, if one exists; otherwise P	$\mathbf{C}[x^h]$, or $\mathbf{C}[P]$
$P = H(x^m)$, $\deg H = 3$, off the aligned-square locus	the same power factor, if it exists; otherwise P	$\mathbf{C}[x^h]$, or $\mathbf{C}[P]$
$P = H(x^{2k}) = (x^{3k} + bx^k)^2 + C$	P, admissible power factors, and $F = x^{3k} + bx^k$ when $F(a) = F(b)$	$\mathbf{C}[P] + \mathbf{1}_{F(a)=F(b)}\mathbf{C}[F] + \mathbf{1}_{a^h=b^h}\mathbf{C}[x^h]$
indecomposable $P = x^4 + \alpha x^2 + \beta x$, with β nonzero	P only	$\mathbf{C}[P]$
$P = (x^4 + \alpha x)^3$, with α nonzero	P, $W_4 = x^4 + \alpha x$, and $W_3 = x^3$, subject to endpoint equality	$\mathbf{C}[P] + \mathbf{1}_{W_4(a)=W_4(b)}\mathbf{C}[W_4] + \mathbf{1}_{a^3=b^3}\mathbf{C}[x^3]$

Here $E_{\min}$ consists of divisibility-minimal elements of

$$E = \{e : e \mid d,\ D_e(a, \alpha) = D_e(b, \alpha)\}. \tag{17.1}$$

In over-power rows, omit the h-term if no divisor of m has equal endpoint values. Lower-rank overlaps use the earliest applicable row.

Proof

Theorem 16.1 says that the solution space is the sum of $\mathbf{C}[W]$ over endpoint-admissible right factors. For x^d, all factors are x^e, $e \mid d$; endpoint equality holds exactly when $h \mid e$, and their algebras lie in $\mathbf{C}[x^h]$.

The Dickson identity

$$D_d(x,\alpha) = D_{d/e}(D_e(x,\alpha),\alpha^e) \tag{17.2}$$

gives its factors. If $e_1 \mid e_2$, then $\mathbf{C}[D_{e_2}] \subseteq \mathbf{C}[D_{e_1}]$; only minimal admissible factors are needed, while incomparable factors must be summed.

For $H(x^m)$, ordinary suffixes are x^e, $e \mid m$, and collapse to the least admissible x^h. If none is admissible, P remains. The sole additional rank-three suffix is

$$H(x^{2k}) - C = (x^{3k} + bx^k)^2. \tag{17.3}$$

It contributes $\mathbf{C}[F]$ exactly when $F(a) = F(b)$.

An indecomposable quartic has no proper nonlinear factor. The exceptional collision has the incomparable factors $x^4 + \alpha x$ and x^3, since

$$(x^4 + \alpha x)^3 = x^3(x^3 + \alpha)^3 = \widetilde{H}(x^3). \tag{17.4}$$

The table follows from Theorem 16.1. □

Corollary 17.2 — Decision procedure

Affine-normalise P, locate its Atlas row, evaluate the listed factors at a, b, and test whether G belongs to the resulting sum of polynomial subalgebras. This is exact decomposition, not moment sampling.

---

18. Laurent phases and arbitrary Laurent amplitudes

Let

$$L(z) = \sum_{i=-m}^{n} c_i z^i, \qquad m, n \geq 1, \tag{18.1}$$

where c_{-m} and c_n are nonzero, and let $A(z)\,dz$ be any Laurent-polynomial one-form. Divide L by c_{-m}. There is a unique formal series $\phi(\tau) = \tau(1 + O(\tau))$ satisfying

$$L(\phi(\tau)) = \tau^{-m}. \tag{18.2}$$

Theorem 18.1 — Laurent local-channel criterion

For every $k \geq 1$,

$$\boxed{\operatorname{Res}_{z=0} L(z)^k A(z)\, dz = [\tau^{mk-1}] A(\phi(\tau))\phi'(\tau).} \tag{18.3}$$

Thus all Laurent moments vanish if and only if all displayed coefficients vanish.

Proof

Formal residues are invariant under $z = \phi(\tau)$. Therefore

$$\operatorname{Res}_{z=0} L^k A\, dz = \operatorname{Res}_{\tau=0} \tau^{-mk} A(\phi(\tau))\phi'(\tau) d\tau,$$

whose τ^{-1} coefficient is (18.3). □

This includes the logarithmic class $z^{-1}dz$.

Corollary 18.2 — Exact Laurent amplitudes

If $A = G'$ for Laurent polynomial G, then

$$\operatorname{Res}_0 L^k\, dG = mk[\tau^{mk}] G(\phi(\tau)). \tag{18.4}$$

If $G = H \circ L$, this vanishes because $G(\phi) = H(\tau^{-m})$. A second mechanism is deck symmetry: if $L = R(z^e)$, excluded residue classes vanish under $z \mapsto \zeta_e z$. For example, $\operatorname{Res}_0 (z^2 + z^{-2})^k dz = 0$ for every k. This corrects the inherited exploratory notebook's undercount for symmetric phases.

---

19. Why the Laurent result stops here

Theorem 18.1 is exact for specified L, A, but it is not a finite classification of all pairs. Pakovich, Pech, and Zvonkin [PPZ10] exhibit an indecomposable Laurent phase whose persistent solution space is a free module of rank five over $\mathbf{C}[L]$. A simple composition-only description is therefore insufficient.

This release does not infer all-k vanishing from a finite zero prefix, promote inherited small-bidegree tables to a theorem, or claim a finite all-Laurent analogue of Theorem 17.1. It supplies the exact coefficient object against which future proposals must be proved.

---

20. Exact computational checks

The implementation compares Krylov reduction of $A, PA, P^2A, \ldots$ with coefficient extraction from $G(\xi(t))$. It stops only when the witnessed support size equals the independently computed cyclic rank. Controls include powers, Dickson phases, affine transport, the rank-four amplitude, and two exceptional zero-cycle centres.

For Laurent inputs, local inverse coefficients are compared with direct Laurent multiplication and residues at every tested moment. Controls include a logarithmic amplitude, deck symmetry, nonvanishing, target scaling, and exact forms.

Ordinary and optimised executions produce byte-identical receipts. Two semantic mutations are rejected after digest recomputation. Sixteen endpoint moments and finite Laurent prefixes remain regressions, not infinite proofs.

21. Assurance boundary and open problems

This manuscript proves the polynomial-amplitude support formula, rank-zero criterion, specified zero-cycle criterion, and arbitrary-Laurent-form local criterion. The endpoint table additionally imports [PM09] and the unrefereed factor classification in Theorem 11.1. It is a complete specialisation in that framework, not an independent reproof of monodromy or Ritt classification.

Next priorities are external specialist review; formalisation of the moment isomorphism; finite structural conditions equivalent to the Laurent test for controlled phases; and classification of amplitudes of bounded induced rank.

Imported classification sources

The all-degree proof imports four classification inputs.

- Behajaina, König, and Neftin [BKN26], Theorems 2.3, 2.4, 2.7, and 3.3, provide the length-two block kernels and characteristic-zero primitive monodromy alternatives. This is a 2026 preprint whose source-side Magma calculations are not independently certified here.
- Burda and Khovanskii [BK16] transmit the Müller primitive-polynomial list and provide a source-to-list comparison.
- Calbetó [C25], Tables 1 and 2, supplies the low-dimensional finite projective-group lists.
- Zieve and Müller [MZ08] supply the modern Ritt-decomposition framework.

The endpoint classification imports the necessary-and-sufficient polynomial moment theorem of Pakovich and Muzychuk [PM09] and uses Pakovich's explicit right-factor refinement [P13]. Álvarez, Bravo, and Mardešić [ABM13] and Pakovich, Pech, and Zvonkin [PPZ10] delimit the zero-cycle and Laurent composition claims.

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Data and code availability

The candidate package contains the component manuscripts, generated combined sources, exact programs, Lean sources, CAS checks, negative controls, frozen receipts, claim ledgers, literature audits, and deterministic rendering tools. No confidential data are used. Generated combined sources identify their builder and can be reproduced from the preserved component files.

Limitations

The all-degree theorem imports named results on primitive polynomial monodromy, finite projective linear groups, and composition-length-two kernels. The package does not formalise those classifications, algebraic function-field integrality, Kummer theory, or Ritt theory in full. The 2026 kernel source is a current preprint and includes source-side Magma computations not independently reproduced here.

The amplitude and Laurent programs verify exact encoded consequences and examples. They do not constitute independent formal proofs of every semantic bridge. Finite zero prefixes are not promoted to persistent Laurent vanishing. Theorem 18.1 is an exact coefficient criterion for a specified input, not a finite normal-form classification of every Laurent pair.

The literature audit used named public databases, open full texts, Crossref, OpenCitations, OpenAlex, Semantic Scholar, zbMATH Open, and AMS Batch MR Lookup. It was not an exhaustive topical MathSciNet search, universal citation-forward closure, non-English full-text search, or unpublished-work survey. No absolute novelty or priority claim is made.

Ethics declaration

This mathematical research uses no human participants, animals, personal data, or sensitive datasets. No ethics approval was required.

Author contributions (CRediT)

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